

COMMON COURSE ASSIGNMENT REPORT

Fundamentals of Data Science (FODS)

A. Assignment Information

Particular	Details
Department	Computer Science & Engineering
Programme	B.Tech CSE AI
Course Code & Course Name	DSA0404 – Fundamentals of Data Science
Academic Year / Batch	2024
Faculty Name	Dr. M. KRISHNARAJ
Assignment Title	University Parking Occupancy Prediction Using Data Science
Date of Issue	02-09-2026
Date of Submission	02-09-2026
Maximum Marks	100
Course Outcome(s)	CO2, CO3, CO5
Bloom's Taxonomy Level	L4 – Analyse
SDG Mapping	SDG 9 & SDG 11
Industry / Societal Relevance	Smart Campus Infrastructure

B. Assignment Problem / Challenge

Develop a data-driven system that analyzes historical university parking occupancy records to identify peak parking hours, predict parking availability, and improve parking space utilization using data science techniques.

Students should:

- Apply data science concepts rather than reproduce theory.
- Identify and analyze the parking occupancy problem.

- Consider technical and societal constraints.
- Develop an analytical solution using modern tools.
- Interpret results and justify engineering decisions.

C. Problem Statement

University campuses experience severe parking congestion during class hours. Students waste valuable time searching for parking spaces while administrators lack analytical insights into occupancy patterns. This project proposes a data science solution to analyze historical parking data, visualize occupancy trends, and predict future parking availability.

D. Requirements and Constraints

Functional Requirements

- Collect parking occupancy data
- Clean missing and duplicate records
- Analyze hourly occupancy
- Generate visual reports
- Predict future occupancy

Technical Constraints

- Historical dataset only
- Python-based implementation
- Limited sample data

User Requirements

- Easy-to-understand reports
- Accurate occupancy prediction
- Graphical visualization

Sustainability

- Reduce congestion
- Save fuel
- Improve campus mobility

E. Student Work

1. Problem Understanding and Formulation

Problem: Predict parking occupancy using historical data.

Expected Outcome: Identify peak hours and improve parking planning.

Available Data: Date, Time, Zone, Total Slots, Occupied Slots.

Assumption: Parking records are accurate.

Constraint: Analysis depends on historical data quality.

2. Application of Course Knowledge

FODS Concept	Application
Data Collection	Parking records
Data Cleaning	Remove missing values
Descriptive Statistics	Occupancy percentage
Visualization	Bar & line charts
Predictive Analysis	Forecast parking demand

3. Solution / Design / Methodology

1. Load parking dataset
2. Clean invalid records
3. Group data by hour
4. Calculate occupancy percentage
5. Generate visualizations
6. Predict future occupancy
7. Prepare analytical report

Algorithm (Pseudo Code):

START

Load Dataset

Clean Data

Group by Hour

Calculate Occupancy %

Generate Graphs

Predict Occupancy

END

4. Use of Modern Tools

Tool	Purpose	Evidence
Python	Programming	Jupyter Notebook
Pandas	Data Cleaning	Processed dataset
NumPy	Statistical Analysis	Occupancy calculations
Matplotlib	Visualization	Bar & Line Charts

5. Results and Validation

Time	Occupied	Occupancy %
8 AM	82	41%
9 AM	128	64%
10 AM	176	88%
11 AM	192	96%
12 PM	181	90%

Validation: The highest parking occupancy occurs at 11:00 AM (96%), satisfying the objective of identifying peak parking demand.

6. Analysis and Engineering Decision

Advantages:

- Peak-hour identification
- Better parking planning

- Reduced search time
- Data-driven decision making

Limitations:

- Historical data only
- No real-time sensors
- Prediction depends on dataset quality

Final Decision: The data science approach was selected because it provides measurable insights through statistical analysis and visualization, making it more effective than manual parking monitoring.

7. Broader Considerations

SDG 11 – Sustainable Cities & Communities: Reduces congestion and improves transportation efficiency.

SDG 9 – Industry, Innovation & Infrastructure: Supports smart campus infrastructure through data-driven parking management.

8. Conclusion

The proposed system successfully analyzes parking occupancy patterns, identifies peak demand, and provides predictive insights using data science. The project achieves the intended course outcomes while improving campus resource utilization.

9. Student Reflection

This assignment helped me apply data collection, preprocessing, visualization, and predictive analysis to solve a real-world campus problem. With additional time, I would integrate IoT parking sensors and machine learning models for real-time occupancy prediction.

10. References

8. Pandas Documentation
9. NumPy User Guide
10. Matplotlib Documentation
11. Python Software Foundation
12. OpenAI ChatGPT (Documentation Assistance)

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course knowledge	20
Solution methodology / implementation	20
Use of modern tools	10
Results & validation	15
Analysis & justification	15
Broader considerations	5
Technical documentation & reflection	5
TOTAL	100

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	CO2	PO1	L3/L4	10
Application of knowledge	CO3	PO2	L3/L4	20
Solution / Implementation	CO5	PO3	L4/L5/L6	20
Modern tool usage	CO5	PO5	L3/L4	10
Validation	CO3	PO4	L4/L5	15
Analysis & justification	CO5	PO2	L4/L5	15
Broader considerations	CO2	PO6	L4/L5	5
Documentation & reflection	CO2	PO10	L3/L4	5

H. Faculty Evaluation & Continuous Improvement

Target CO Attainment: _____

Actual CO Attainment: _____

Faculty Analysis

Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action: